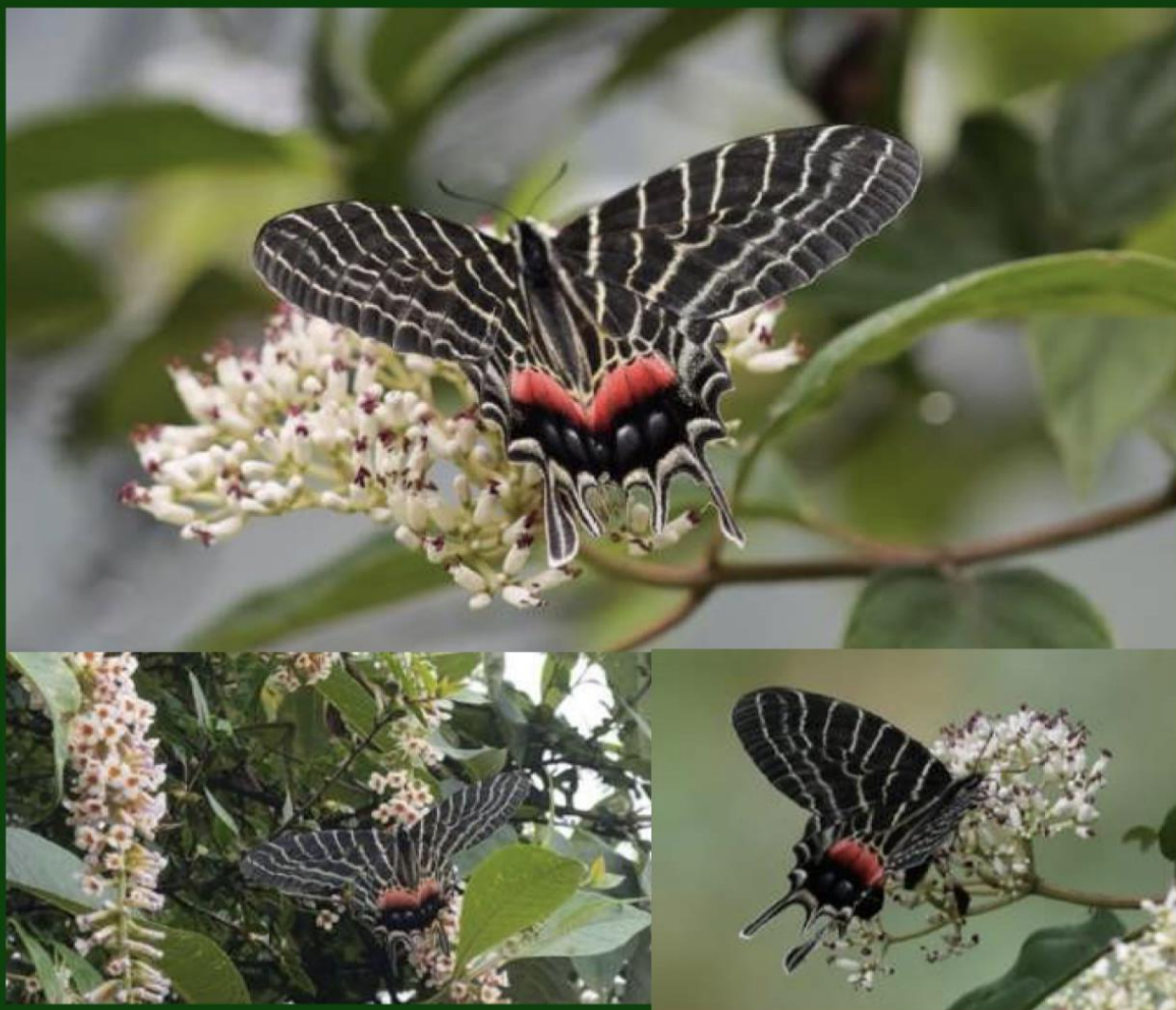




LUDLOW'S BHUTAN SWALLOWTAIL

Bhutanitis ludlowi Gabriel

CONSERVATION ACTION PLAN FOR BHUTAN 2025 - 2034



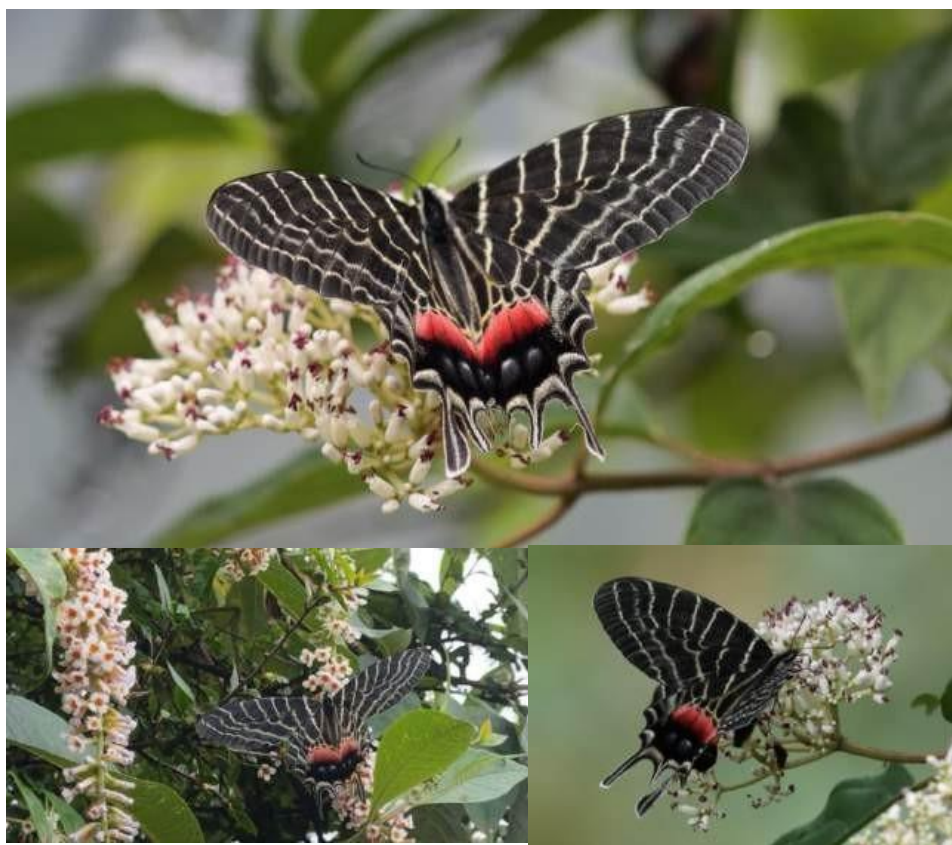
Bumdeling Wildlife Sanctuary

LUDLOW'S BHUTAN SWALLOWTAIL

Bhutanitis ludlowi Gabriel

CONSERVATION ACTION PLAN

(2025-2034)



Bumdeling Wildlife Sanctuary
Department of Forests and Park Services
Ministry of Energy and Natural Resources
Royal Government of Bhutan



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MINISTRY OF ENERGY AND NATURAL RESOURCES
DEPARTMENT OF FORESTS AND PARK SERVICES



DIRECTOR

FOREWORD

The Department of Forests and Park Services is committed to managing and conserving Bhutan's rich biodiversity. Our primary focus is the protection of natural resources and the conservation of threatened flora and fauna. Through continuous efforts, we have established Bhutan as a global leader in environmental conservation in the 21st century.

Conservation action plans serve as vital road maps for protecting species facing threats and challenges. Over the past four to five years, the department has made significant progress in developing action plans for key flagship species, including the Tiger, Snow Leopard, Red Panda, Rufous-necked Hornbill, Black-necked Crane, and White-bellied Heron. Recognizing the importance of butterfly conservation, we are pleased to introduce Bhutan's first-ever conservation action plan for the rare Ludlow's Bhutan Swallowtail (*Bhutanitis ludlowi* Gabriel).

This species was rediscovered in Tobrang, Trashiyangtse, in 2009 and subsequently designated as Bhutan's National Butterfly in 2012. In 2023, it was officially listed under Schedule II of the Forest and Nature Conservation Act of Bhutan. The swallowtail's habitat overlaps with human settlements, making it vulnerable to various threats that necessitate targeted conservation measures. This action plan will be implemented by field offices under the Department of Forests and Park Services, and we look forward to continued collaboration with NGOs, institutions, and conservation projects. Through our collective efforts, we can ensure the long-term survival of this symbolic butterfly while promoting a harmonious coexistence between wildlife and communities.

Lastly, I extend my sincere appreciation to my colleagues at Bumdeling Wildlife Sanctuary, the Nature Conservation Division, and the Ugyen Wangchuck Institute for Forestry Research and Training for their dedication in developing this crucial document. I wish them success in its implementation.

Karma Tenzin
Director

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The Bumdeling Wildlife Sanctuary, Trashiyangtse under the Department of Forests and Park Services would like to acknowledge the support rendered by Nature Conservation Division and Ugyen Wangchuck Institute for Forest Research and Training (UWIFoRT) as this Conservation Action Plan is an outcome of our collective effort and contribution.

The management is highly grateful to Ludlow's Habitat Assessment Survey team, 2023 for collecting the invaluable field data for producing this document, despite downpours during the field survey.

This Action Plan would not have been materialized without the funding support from Global Environment Facility (GEF), for which the management would like to remain highly indebted for their continued support in conservation of biodiversity and climate change.

Lastly, we look forward to continued support from our partner institutions and donors towards implementing the actions prescribed in this plan, so that Ludlow's Bhutan Swallowtail thrive in its habitat through our collective conservation efforts.

ACRONYMS

BFL	Bhutan For Life
BT FEC	Bhutan Trust Fund for Environmental Conservation
BWS	Bumdeling Wildlife Sanctuary
CFMG	Community Forest Management Group
CITES	Convention on International Trade on Endangered Species
DAO	District Agriculture Office
DLO	District Livestock Office
DoFPS	Department of Forests and Park Services
FNCA	Forest and Nature Conservation Act
FNCRR	Forest and Nature Conservation Rules and Regulations
GEF	Global Environment Fund
IUCN	International Union for Conservation of Nature
m.a.s.l	meter above sea level
NCD	Nature Conservation Division
RGoB	Royal Government of Bhutan
SMART	Spatial Monitoring and Reporting System
SRFL	State Reserved Forest Land
UNDP	United Nation Development Programme
UWIFoRT	Ugyen Wangchuck Institute for Forest Research and Training
WWF	World Wildlife Fund

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EXECUTIVE SUMMARY

Ludlow's Bhutan Swallowtail (*Bhutanitis ludlowi*) is an endangered butterfly native to Bhutan and the Arunachal Pradesh region of India. It was first discovered in the 1930s in Tobrang, Bhutan, by English botanist Frank Ludlow. Owing to its rarity, this species has been designated as Bhutan's National Butterfly and is listed under Schedule II of the Forest and Nature Conservation Act of Bhutan, 2023.

Ludlow's Bhutan Swallowtail typically inhabits elevations ranging from 1,500 to 3,000 meters above sea level and is most active during daylight hours. It feeds on the nectar of various plant species and lays its eggs on *Aristolochia griffithii*. The species faces significant threats from uncontrolled grazing, resource extraction, agricultural practices, and developmental activities, all of which contribute to habitat degradation—particularly because its habitat lies close to human settlements. To ensure the conservation of Ludlow's Bhutan Swallowtail and its environment, a comprehensive action plan has been formulated, centered around four primary objectives:

- Generate baseline data on Ludlow's abundance, conservation threats and social demographic status within Ludlow's potential habitats.
- Conserve Ludlow's habitats based on scientific habitat management.
- Advocate local communities on Ludlow conservation and involve in participatory action planning, implementation and monitoring processes.
- Promote community engagement in Ludlow's conservation and support nature-based enterprises, eco-tourism, and other livelihood programs.

The ten-year plan has an estimated budget of Nu. 18.1 million, which will be financed through the Global Environment Fund (GEF-7) and Bhutan for Life initiatives. Its implementation will be monitored and evaluated annually, with mid-term reviews to ensure progress and effectiveness. Ultimately, the plan aims to achieve sustainable habitat conservation and maintain a healthy population of Ludlow's Bhutan Swallowtail through active engagement of local communities and regional stakeholders.

CHAPTER 01: BACKGROUND

1.1. Discovery and Distribution of *Bhutanitis ludlowi*

Ludlow's Bhutan Swallowtail, *Bhutanitis ludlowi* Gabriel (Lepidoptera: Papilionidae), was first discovered in a small area of Tobrang, upstream of the Trashiyangtse Valley, by two English botanists, Frank Ludlow (1885–1972) and George Sheriff (1898–1967), during a botanical expedition (Stearn, 1976). However, it took eight years to confirm and formally describe the species, which was done by Gabriel in 1942, naming it in honor of Frank Ludlow. After a gap of 79 years, the butterfly was rediscovered in 2009 by Karma Wangdi, who is currently serving as a Research Assistant at the Ugyen Wangchuck Institute for Forest Research and Training in Bumthang (Harada *et al.*, 2012). Initially considered endemic to Tobrang in Bhutan's Trashiyangtse District, the species was later recorded in 2012 in Senge village, located within Eaglenest Wildlife Sanctuary in Arunachal Pradesh, India (Singh and Sondhi, 2016; Sondhi and Kunte, 2016; Varshney and Smetacek, 2015 eds.).

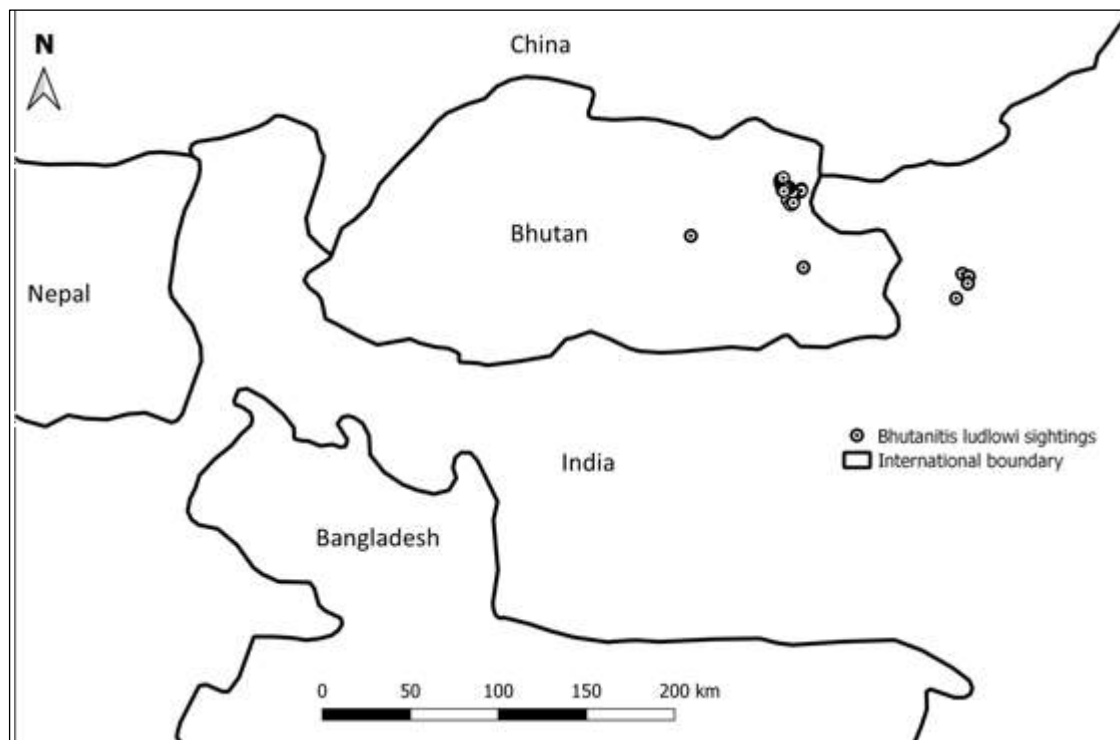


Figure 01. *Distribution of Bhutanitis ludlowi*

Bhutanitis ludlowi is distributed across parts of Bhutan and India. In Bhutan, it has been recorded in the Trashiyangtse District (Bumdeling & Tobrang), Trashigang District (Khaling & Shoskum), and Bumthang District (Ura & Khandopang). In India, its presence has been noted in Arunachal Pradesh, specifically in Senge and Lama Camp.

Distribution records from Arunachal Pradesh are sourced from the Butterflies of India website (<https://www.ifoundbutterflies.org/bhutanitis-ludlowi>). Additionally, a specimen was collected from Yunnan Province, China, in 1992, although its current presence in China remains uncertain. This indicates that *Bhutanitis ludlowi* may have a broader range than previously assumed. The species is typically found at elevations between 2,200 and 2,500 meters above sea level (Collins and Morris, 1985; Harada et al., 2012).

1.2. Research and conservation efforts of *Bhutanitis ludlowi* in the range countries

Due to its limited distribution in restricted areas, rarity, ongoing habitat loss and disturbances throughout its range, and insufficient data on population trends, *Bhutanitis ludlowi* has been classified as an endangered species by the International Union for Conservation of Nature (IUCN) (Fernando et al., 2019). Owing to its rarity, it is also listed under Appendix II of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES). Despite its current presence within protected areas in both Bhutan and India, the long-term survival of its population remains uncertain. Therefore, enhanced conservation efforts and systematic monitoring through empirical studies are essential to ensure effective management and protection of this rare species.

Following the rediscovery of *Bhutanitis ludlowi*, a collaborative research effort between Bhutan and Japan was promptly initiated. This study (Harada et al., 2012) clarified that the butterfly was not *Bhutanitis lidderdalii* Atkinson (Bhutan Glory), as previously believed. The researchers also identified the larval host plant as *Aristolochia griffithii* Hook.f. & Thomson ex Duch. Moreover, the team confirmed that *Bhutanitis ludlowi* is an extremely rare species, endemic to Bhutan. This recognition led to its designation as Bhutan's National Butterfly in 2012 (Singh and Chib, 2015).

According to the Forest and Nature Conservation Act (FNCA) 2023, *Bhutanitis ludlowi*, along with *Bhutanitis lidderdalii* and *Megalestes gyalsey*, is listed under Schedule II and granted the highest level of legal protection.

More recently, a study on the complete ex-situ life cycle of *Bhutanitis ludlowi* revealed that it takes approximately 365–367 days to complete its life cycle, with the longest phase being pupal development (Dendup et al., 2023). The study also identified natural predators during the egg and larval stages, including spiders, wasps, praying mantises, snails, and ants. However, the species defends itself by releasing a repellent toxic odor and using camouflage (Dendup et al., 2023).

In addition, a rapid habitat assessment survey conducted by Bumdeling Wildlife Sanctuary (BWS) in collaboration with the Ugyen Wangchuck Institute for Forest Research and Training (UWIFoRT) found that *Bhutanitis ludlowi* is typically found along forest edges and in forest-pasture or meadow interfaces within cool broadleaf forest zones.

1.3. Behaviour and Ecology

Adult *Bhutanitis ludlowi* are typically active on sunny days but may also fly during cloudy conditions if temperatures are favorable (Harada *et al.*, 2012). Field observations have shown that a higher number of individuals take flights above tall trees during clear, sunny weather. These butterflies are commonly seen within the canopies of *Quercus griffithii* and *Alnus nepalensis*, and often rest or roost beneath the leaves of these trees after feeding. Detailed observations indicate that *Bhutanitis ludlowi* emerges from the pupal stage as a fully developed adult in July and disappears by October. Additionally, Harada *et al.* (2012) reported that male butterflies begin to appear in early August, while females emerge from mid-August onward (Figure 02).



Figure 02. The life cycle of *Bhutanitis ludlowi* illustrates the main developmental stages completed over the course of a year. © Dendup *et al.* (2022)

1.4. Current distribution and habitat range of *Bhutanitis ludlowi* in Bhutan

A survey conducted in 2021 by BWS in collaboration with UWIFoRT validated the spatial information of its occurrence substantiated by population estimates in the area through Mark and recapture method. It was found that the elevational range of butterfly occurrence extended to 2900 m from 2500 m.a.s.l previously reported as upper limit (Unpublished report). Currently, *Bhutanitis ludlowi* has been sighted in areas such as Pelrigonpa, Cheng, Barigang, Phomodung, Tarphe-Tobrang, Rigsum Gonpa, Pangkhar, Pangliwog, Laptsa, Dechen Phodrang, and Tokadung Kharungla. Recently, sightings from Khandupang in Ura, Bumthang (Wangmo, 2020; Dorji and Dorji, K.T, 2021) and Shoskum Hill, Khaling Gewog in Trashigang were reported (Kipchu and Darjay, 2021; Dorji and Dorji, K.T, 2021).

It is also likely to be found in other yet unexplored areas of Bumdeling that share similar climatic and ecological conditions. These locations fall within an elevation range of 2200 to 2900 meters above sea level, with the highest recorded sighting at 2900 m.a.s.l. A study by Choidup (2016) on the distribution of the butterfly's host plant revealed that the greatest abundance of *Aristolochia griffithii*, the primary host plant, was found between 2300 and 2400 m.a.s.l. Sightings of *Bhutanitis ludlowi* are consistently associated with the presence of its host plant; however, the availability of open areas such as forest-grassland and farmland-forest edges also supports the butterfly's feeding and flight activities.

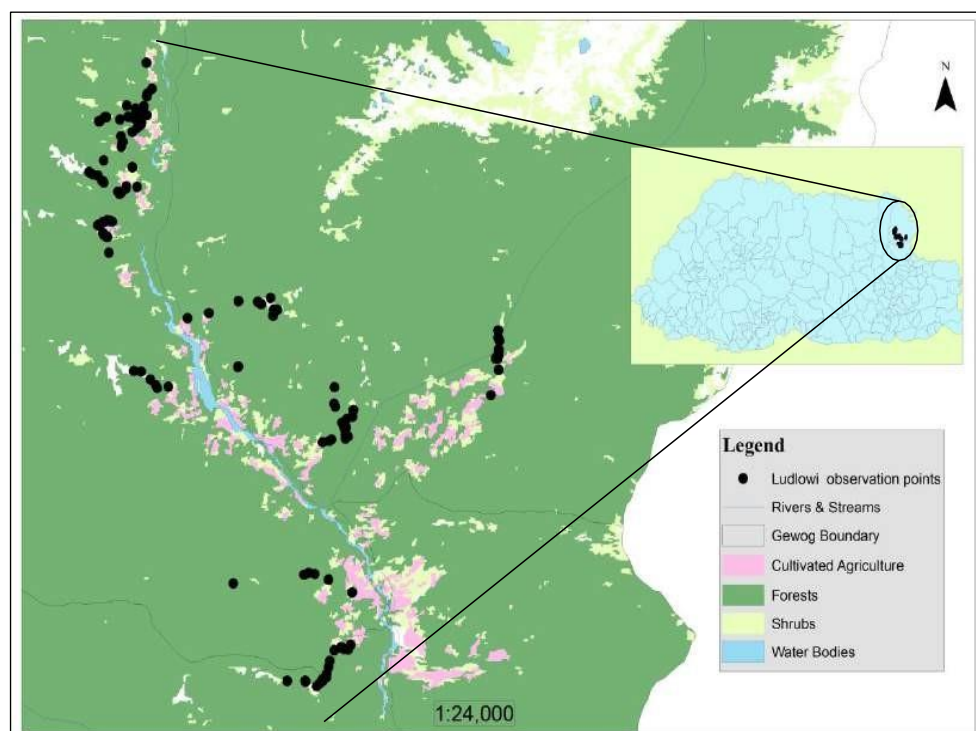


Figure 03. Distribution of *Bhutanitis ludlowi* in Bumdeling

1.5. Feeding and host plants

Aristolochia griffithii is the preferred host plant for *Bhutanitis ludlowi*, where females lay their eggs and the larvae feed and develop (Choidup, 2016). Species of *Aristolochia* play a crucial role in the butterfly's life cycle, serving as the primary plant for egg-laying, larval feeding, and the development through various pupal instars (Dendup et al., 2023) (Figure 4(a) & (b)). The foliage of *Aristolochia* contains toxic compounds that deter most predators; as a result, the larvae feeding on these leaves are typically avoided by birds and other predators.

While adult butterflies feed on nectar from a variety of plants (David and Robert, 1987), in Bumdeling, their primary nectar source is *Viburnum cylindricum*, along with *Aconogonum molle* and *Buddleja asiatica* (Choden, 2011; Harada et al., 2012) (Figure 4(c)). This reliance may be due to the limited availability of flowering plants that bloom during the active season of *Bhutanitis ludlowi*.

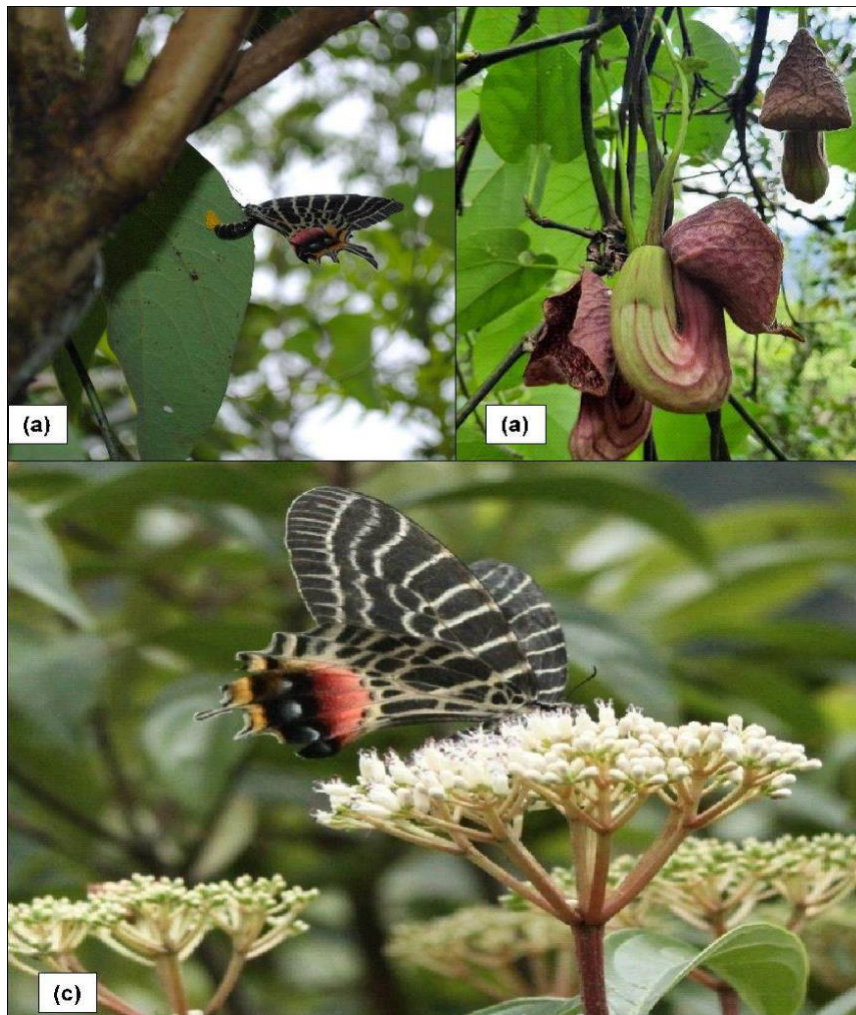


Figure 04 (a) Female Ludlowi laying eggs on the leaf of *Aristolochia griffithii*, (b) *Aristolochia* plant and flower and (c) Adult Ludlowi feeding on *Viburnum cylindricum* flower.

1.6. Adult Ludlowi counts in the habitat

Eight transect lines were predetermined and laid out in the prime habitat of Ludlowi in Bumdeling for carrying out the survey of *Bhutanitis ludlowi* in 2023 (Figure 05). Only one time survey was conducted to collect data using Eppicollect5 app to know the encounter rate for each transects. The total distance covered by foot transect was 37.588 km and the study transects passes through different habitat regime used by Ludlowi such as open forested area, Shrub land, grasslands, Cool Broadleaved Forest with tall Oak and Alnus trees. Details of transects are given in Table 01.

The survey result showed that the highest individual sighting of *Bhutanitis ludlowi* was along Khayateng-Cheng transect (N=60) followed by Tobrang-Poktorla (N=40), Pangkhar-Shaliteng (N=38), Betsamang-Shawangla (N=22) and the lowest count in along Khayateng-Tobrang (N=17). A high encounter rate was found in Pangkhar-Shaliteng and Tobrang-Poktorla transect, probably because the transect was along the ridge with more altitudinal variation and passes through settlement, open grasslands, shrub forest, thick and high forest with tall trees. On the other hand, the low encounter rate was found in Dungzam-Pelrigonpa and Betsamang-Shawangla.

Table 01: Encounter rate estimation for 8 survey transects

Survey Transect Name	Distance covered (km)	Total count	Encounter Rate (per km)
Khayateng-Cheng	7.2	60	8.39
Pangkhar-Shaliteng	2.7	38	14.24
Tobrang-Poktorla	3.5	40	11.39
Khayateng-Tobrang	2.8	17	5.99
Dungzam-Pelrigonpa via. Rigsum Gonpa	10.2	21	2.07
Betsamang-Shawangla	4.9	22	4.47
Dechenphodrang-Brema ju	2.7	19	6.96
Gangkhardung-Yulapeng	3.6	27	7.50

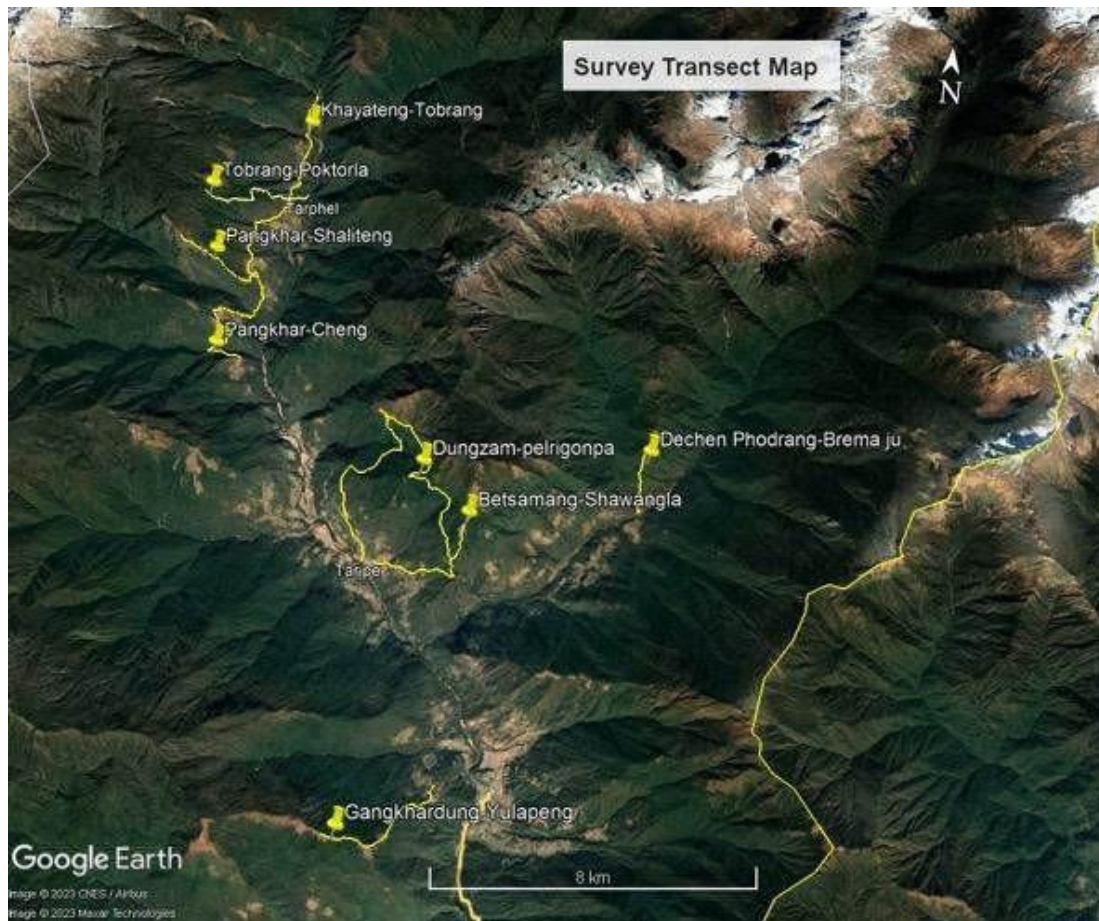


Figure 05: *Ludlowi* survey transects in Bumdeling

1.7. National Policies and Conservation efforts in Bhutan

Environmental conservation is one of the four pillars of Gross National Happiness envisioned by His Majesty the Fourth Druk Gyalpo. Article 5 of the Constitution mandates the government to recognize every Bhutanese as a ‘trustee of the Kingdom’s natural resources and environment’. The Forest and Nature Conservation Acts and Rule of Bhutan also accords the highest protection status to it after coming into existence since the rediscovery of the species in 2009 from Tobrang in Trashiyangtse. The *Bhutanitis ludlowi* was endorsed and officially declared as Bhutans’ National Butterfly during the 123rd sitting of Cabinet in 2012 (Singh & Chib, 2015). It is also listed under the Schedule II of Forest and Nature Conservation Act (FNCA), 2023. The Forest and Nature Conservation Rules and Regulations (FNCRR) of Bhutan 2023 testify a fine of Nu. 5000.00 (Nu. Five thousand) for killing, injuring and destroying and Nu. 2500 (Nu. Two thousand five hundred) for capturing, collecting, breeding and raising.

CHAPTER 02: THREAT AND CHALLENGES

2.1. Introduction to threats and challenges

Habitat loss and environmental degradation are the major threats for *Bhutanitis ludlowi* in its habitat, like many other vulnerable and endangered species. These threats are intensified by human disturbances and ecological changes driven by natural forces. The pressure is particularly severe in both existing and potential habitats, where butterfly feeding and host plants have been significantly affected by uncontrolled grazing, agricultural activities, resource extraction, developmental works, and the impacts of climate change.

A rapid threat assessment survey was carried out by laying 66 plots (10 m x10 m) along 11 predetermined transects to validate the information on the threats. The result of the assessment showed that un-controlled grazing pressure and extraction of resources as **‘High’** and other threats as **‘Low’** rating. The identified threats and challenges were also analysed in *Miradi – Adaptive management software for conservation projects* and found that resource extraction and climate change rated as **‘High’**, un-controlled grazing pressure and limited conservation efforts as **‘Medium’**. The remaining threats such as farming practices, developmental activities, limited research and information gap, limited community participation and awareness, limited coordination amongst stakeholders as **‘Low’** (Table 02).

Table 02: Overall threats and challenges rating in *Miradi*

Threats and Challenges	Objective 1: Generate baseline data on Ludlow's abundance, conservation threats and social demographic status within the potential habitats.	Objective 2: Conserve Ludlow's habitats based on scientific habitat management.	Objective 3: Advocate local communities on Ludlow conservation and involve in participatory action planning, implementation and monitoring processes.	Objective 4: Promote community engagement in Ludlow's conservation and support nature-based enterprises, eco-tourism, and other livelihood programs.	Summary Threat Ranking	
Un-controlled grazing pressure	Low	High	Low		Medium	
Resource extraction	High	High			High	
Farming practices	Medium	Low		Low	Low	
Developmental Activities		Low	Low		Low	
Climate Change	High	High	Low		High	
Limited research & information gap	Low	Medium			Low	
Limited community participation & awareness			Low	Low	Low	
Limited coordination amongst stakeholder			Low	Low	Low	
Limited conservation efforts	Low	High			Medium	
Summary Target Ratings	High	High	Low	Low	Overall Project Rating	High

Threat rating criteria in Miradi			
Rank	Scope	Severity	Irreversibility
Very High	4	4	4
High	3	3	3
Medium	2	2	2
Low	1	1	1

2.2. Threat description

2.2.1. Uncontrolled Grazing pressure

The habitats of *Bhutanitis ludlowi* are generally located near human settlements and are easily accessible to both people and livestock. Consequently, the butterfly's feeding and host plants are often unintentionally cut by herders and are subject to browsing and trampling by livestock, even though these plants are largely unpalatable to them. A study by Choidup (2016) reported that grazing pressure has significantly affected the distribution and abundance of *Aristolochia griffithii* across the entire Bumdeling habitat. Grazing was identified as the most prominent disturbance, followed by plant cutting, lopping, and tree felling. Grazing pressure ranked highest due to the near year-round presence of cattle in the habitat areas where local communities reside.

2.2.1. Resource Extraction

Intensive resource extraction poses a significant threat to the conservation of *Bhutanitis ludlowi* across all its habitats. The lopping and cutting of trees for domestic purposes and livestock fodder have led to habitat degradation. As the butterfly's habitats are easily accessible, local residents frequently visit these areas to collect posts and poles at their convenience. In doing so, they often cut down trees that serve as support structures for the host plants on which the butterfly depends. Additionally, timber extraction and the collection of leaf litter further compromise habitat integrity. To address these challenges, advocacy programs highlighting the importance of conservation, along with education on forestry regulations and the consequences of illegal activities, could play a key role in mitigating these threats.

2.2.2. Farming practices

The practice of slash and burn for growing cereal crops by residents/ communities are evident in their feeding localities. Rural communities in Bumdeling heavily rely on millet and paddy for their subsistence, and to enhance yields and expand cultivation, they frequently shift farming

sites, using slash-and-burn techniques to prepare land for sowing cereal crops (Figure 08). While shifting cultivation was widely practiced in the past, it has been largely discontinued in State Reserved Forest Land (SRFL) due to awareness and advocacy efforts in accordance with existing regulations. However, farmers continue to clear and burn land along forest edges, with formal approval from the Department of Forests and Park Services (DoFPS).



Figure 08. *Slash and burn at the edge of forest land*

2.2.1. Developmental activities

Developmental activities, such as roads construction, electricity line installation, and construction of infrastructure, pose a significant threat to Ludlowi habitat management. While these activities aim to improve the livelihood of rural communities, they often result in extensive environmental damage, including habitat destruction and depletion of natural resources. In the process of cutting down trees and clearing vegetation for road alignment and electricity transmission lines, the preferred feeding and host plants are cut down further impacting to the conservation of Ludlowi.

2.2.2. Climate Change

Climate change poses a significant threat to the survival of *Bhutanitis ludlowi*, warranting detailed research. Observations indicate that the species' habitat experiences heavy and erratic rainfall during the summer months, coinciding with the emergence period of Ludlowi. At times, rainfall persists for several consecutive days, hindering the butterflies' ability to fly and feed, which may lead to starvation-related mortality. Additionally, high humidity levels in the area could adversely affect egg development, potentially preventing full germination. Field observations revealed that Ludlowi are rarely seen during periods of heavy rainfall and elevated humidity, whereas more individuals are active and feeding during sunny weather when humidity levels are lower.

2.3. Challenges

2.3.1. Limited research and information gap

Bhutanitis ludlowi, an endemic species of the Himalayan region, is confined to a small geographic range across parts of Bhutan and India. However, there is a lack of comprehensive research and data on the species in this region, highlighting the need for in-depth studies to understand its long-term viability. Critical information regarding its interactions with host and nectar plants, habitat preferences, and population dynamics remains scarce, making it challenging to develop and implement effective conservation and management strategies.

2.3.2. Limited community awareness and participation

The species remained largely unknown to local communities until its rediscovery in 2009. Following its public recognition and in light of its rarity and status as an endemic species to Bhutan, *Bhutanitis ludlowi* was promptly declared as the National Butterfly of Bhutan in 2012, reinforcing its conservation significance. Although the butterfly shares its habitat—land, forests, and meadows—with the nearby residents, awareness and understanding of the species and its conservation needs remain limited among the local population. Consequently, community engagement and stakeholder participation in conservation efforts have been minimal.

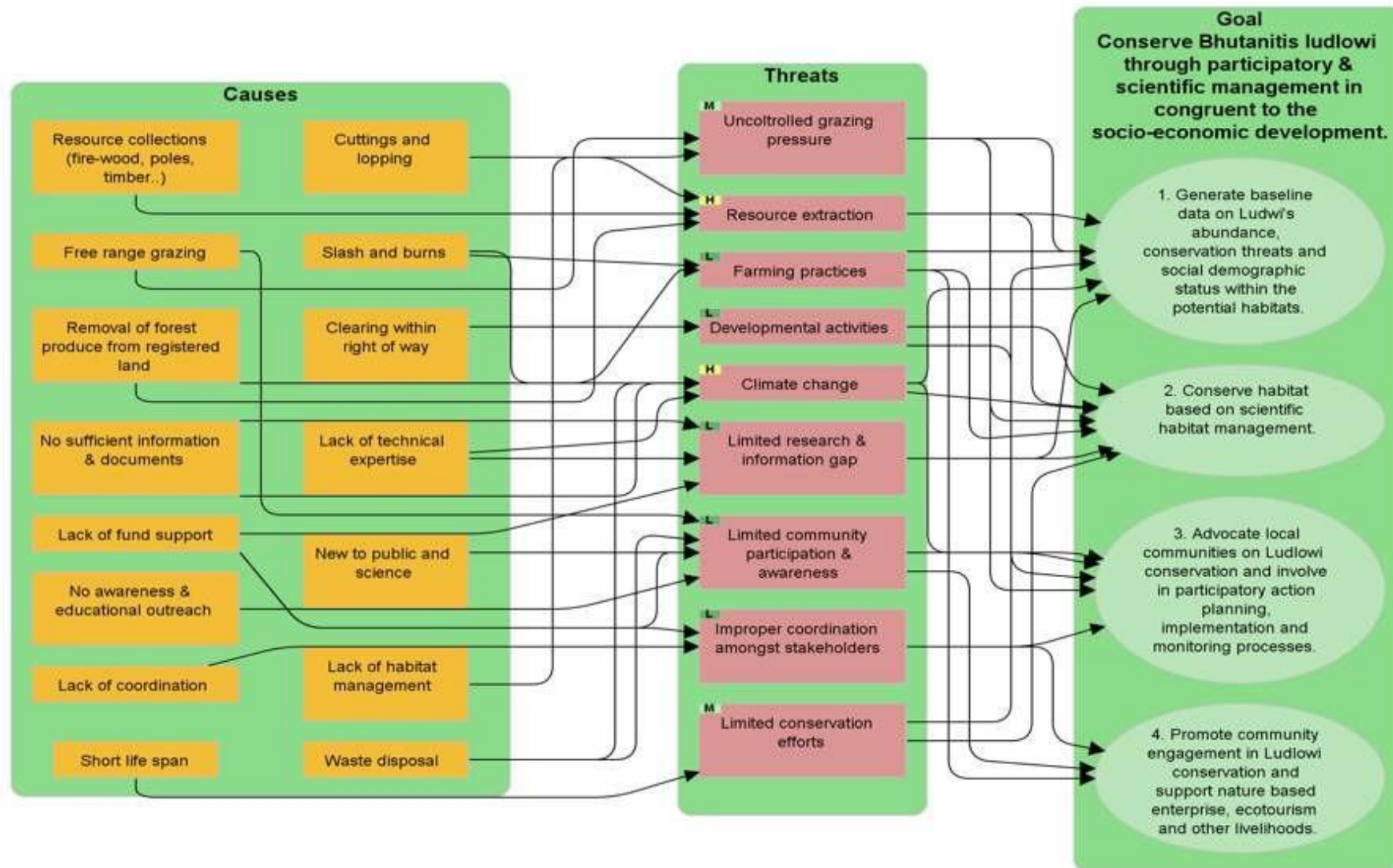
2.3.3. Improper coordination amongst stakeholders

For the conservation of this endemic species, it is very important to have proper planning and coordination amongst the organisations DoFPS, institutions (UWIFoRT) and Local Governments (Dzongkhag and Gewog). Like other conservation activities, one of the challenges pointed out for the conservation of Ludlowi is the existence of weak coordination amongst the stakeholders. If these challenges are not addressed promptly, it may result in duplication of conservation efforts and potential conflicts among agencies involved in habitat management.

2.3.4. Limited conservation efforts

To date, limited conservation efforts have been undertaken to protect *Bhutanitis ludlowi* and its habitat, apart from monitoring studies initiated after its rediscovery in 2009. Bumdeling Wildlife Sanctuary, as the field conservation office under the Department of Forests and Park Services (DoFPS), has identified only a few Ludlowi hotspot areas, conducted habitat mapping, and implemented monitoring activities in collaboration with UWIFoRT, supported by funding from the GEF project and other international donors. Park officials have made small-scale efforts to educate and raise awareness among local communities about the species. However, these measures remain insufficient. There is an urgent need to conduct mass awareness campaigns and build the capacity of field staff. To enhance awareness and gain the support of stakeholders, local communities, and local governments, it is essential to implement a comprehensive advocacy program.

Figure 09: Conceptual Model



CHAPTER 03: ACTION PLAN

3.1. Vision

- ✓ Secure healthy population of Ludlow's Bhutan Swallowtail *Bhutanitis ludlowi* in its natural prime habitats.

3.2. Goal

- ✓ To conserve *Bhutanitis ludlowi* through participatory and scientific management in congruent to the socio-economic development.

3.3. Objectives, outputs and actions

Objective 1: Generate baseline data on *Bhutanitis ludlowi* abundance, conservation threats and social demographic status within the potential habitats.

Rationale:

*In order to ensure the conservation of *Bhutanitis ludlowi* habitat in Bumdeling and even beyond, it is necessary to collect adequate baseline information and data on the species itself, habitat and ecology. It can be done basically through surveys and onsite field assessments.*

The objective 1 focuses on habitat mapping efforts and habitat assessments. The regular monitoring of habitats and population surveys will help monitor the population trends in its habitat range.

Output 1.1: *Bhutanitis ludlowi* habitats mapped and assessed.

Action 1.1.1. Carry out *Bhutanitis ludlowi* distribution and habitat mapping.

Action 1.1.2. Conduct ecological assessment in *Bhutanitis ludlowi* habitats.

Action 1.1.3. Carry out annual population monitoring of *Bhutanitis ludlowi* and its habitat in identified permanent transect.

Action 1.1.4. Map all the communities in the habitat and study their knowledge and opinion on conservation of *Bhutanitis ludlowi*.

Output 1.2: Conservation threats identified and mapped.

Action 1.2.1. Carry out grazing mapping and impact assessment.

Action 1.2.2 Carry out impact assessment through farming practices

Action 1.2.3 Carry out impact assessment through resources allocation

Objective 2: Conserve Ludlow's habitats based on scientific habitat management.

Rationale:

The threats and disturbances to the habitat will lead to habitat degradation and a decline in the population within the area. Threats such as uncontrolled grazing, resource extraction, developmental activities and farming practices contribute to habitat degradation. If conservation actions are not implemented promptly, the population will gradually decline and potentially become extinct in the area. Therefore, the main objective is to address these issues and enhance the habitat's suitability and potential through activities such as implementing enrichment plantations, developing site management guidelines, and constructing fences in certain hotspots to support the species' lifecycle.

Output 2.1. Habitats of *Bhutanitis ludlowi* enhanced through plantation of host and feeding plants

Action 2.1.1. Upscale the existing nursery to supply enough host and feeding plant seedlings.

Action 2.1.2: Extend the habitat of *Bhutanitis ludlowi* through creation/ enrichment plantation of host and feeding plants (5 sites)

Action 2.1.3. Develop Mud Flakes for butterflies in the habitat (5 sites)

Output 2.2. Ludlowi habitats protected

Action 2.2.1: Protect prime *Bhutanitis ludlowi* habitat through natural fencing at some critical habitat with lesser disturbances to other wildlife species

Action 2.2.2: Ensure proper management of waste in the habitat and along the trails.

Output 2.3 Reduce threats through periodic SMART patrol

Action 2.3.1. Carryout SMART patrol on regular basis

Objective 3: Advocate local communities on Ludlow conservation and involve in participatory action planning, implementation and monitoring processes.

Rationale:

The local community and stakeholders residing in and around the habitat area need to be well-informed about the nationally significant species and its habitat. Though it is declared as the National Butterfly of Bhutan, a majority of the people and community in the area remain unaware of its existence. There is also a lack of knowledge among stakeholders regarding the species behaviour, habitat requirements, as well as its feeding and host plants. Hence, it is crucial to establish effective communication and collaboration among various stakeholder groups within the butterfly's habitat range, forming a comprehensive nation-wide conservation strategy.

This objective recognizes that conservation of the species cannot be accomplished in isolation but depends on the active involvement of a diverse and empowered network of stakeholders. This can be achieved through enhanced communication channels, robust peer-to-peer and institutional networking, and support for capacity building initiatives.

Output 3.1. Conservation awareness and participatory planning on Ludlowi conservation enhanced

Action 3.1.1: Develop Information, education and communication materials on *Bhutanitis ludlowi* for BWS information centre.

Action 3.1.2. Produce video documentary of *Bhutanitis ludlowi* life cycle.

Action 3.1.3: Conduct awareness, advocacy and education programs at all levels of stakeholders (monastic institutions, schools, youths and the general public).

Action 3.1.4: Capacity development of park staff on butterfly ecology and survey techniques.

Action 3.1.5. Organize butterfly spotting programs for schools and communities.

Action 3.1.6: Involve stakeholders in planning, reviews and evaluation of *Bhutanitis ludlowi* conservation action plan.

Objective 4: Promote community engagement in *Bhutanitis ludlowi* conservation and support nature-based enterprises, eco-tourism, and other livelihood programs.

Rationale:

Conservation efforts for Bhutanitis ludlowi cannot be sustainably governed without support and involvement of local communities. Therefore, conservation programs and management strategies must be linked with investment in local livelihoods, community interests, and conservation values.

This objective recognizes that community engagement and efforts to support sustainable livelihood opportunities are crucial to ensuring the longevity of Ludlowi conservation in Bumdeling. In order to achieve the target objectives, the broader community must consider local development goals that prioritize livelihood generating activities that either do-no-harm or directly benefit Ludlowi.

Output 4.1. Strengthened community participation and enhance community resilience through diversification of livelihood activities

Action 4.1.1. Initiate ecotourism focused on Ludlowi to enhance awareness and conservation

Action 4.1.2: Provide trainings to local youths on nature guide

Action 4.1.3 Procure equipment for local nature guides

Activity 4.1.4. Support alternative fencing materials to minimize felling and collection of host tree for *Aristolochia griffithii*

CHAPTER 04: IMPLEMENTATION AND MONITORING PLAN

4.1. Institutional arrangement

The DoFPS within the Ministry of Energy and Natural Resources (MoENR) is the institution under the RGoB which is mandated for conservation of biodiversity. As such, DoFPS will be the governing body that coordinates the implementation of the *Bhutanitis ludlowi* conservation action plan in Bhutan. Within the DoFPS, the plan implementation will be coordinated by the Nature Conservation Division in collaboration with respective field divisions/ protected areas and other relevant agencies. However, the field divisions and protected areas in which the Ludlowi habitats fall will take the lead role in implementing the activities projected in the action plan in close collaboration with stakeholders beyond the DoFPS. Besides surveys and assessment, the implementation of the long-term research and development component of the plan will be led by UWIFoRT.

The field divisions and protected areas will implement the activities based on its conservation mandates as per the conservation action plan/ operation plan and budget allocation. Upon contribution from international organizations and institutions on Ludlowi conservation and knowledge generation, experiences sharing will be conducted as and required amongst Non-Governmental Organization, institutions and experts

4.2. Work plan and budget

The total estimated cost required to implement this action plan for the duration of ten years is Nu. 18.1 million (2024-2033) as reflected in the logical framework (Table 03). For financing and implementation of this action plan, the funding support will be explored from Bhutan For Life (BFL) Project, Bhutan Trust Fund for Environmental Conservation (BT FEC), World Wildlife Fund (WWF), United Nation Development Programme (UNDP- GEF-Project), Bhutan Foundation and other international donors and/or RGoB.

Table 03. Logical framework

Vision: - Secured habitat for the healthy population of Ludlow's Bhutan Swallowtail <i>Bhutanitis ludlowi</i> in Bhutan.											
Goal: - To conserve Ludlow's Bhutan Swallowtail <i>Bhutanitis ludlowi</i> and its habitat.											
Objective/ Output and Action	Budget estimate (Nu.)	Timeline									
		Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10
Objective 1: Generate baseline data on Ludlow's abundance, conservation threats and social demographic status within the Ludlow's potential habitats.											
Output 1.1: Ludlow habitats mapped and assessed.											
Action 1.1.1: Carry out distribution and habitat mapping	0.15										
Action 1.1.2: Conduct ecological assessment of Ludlowi habitats	0.2										
Action 1.1.3. Carry out annual population monitoring of Ludlowi and its habitat in identified permanent transect.	1										
Action 1.1.4 Map all the communities in the habitat and study their knowledge and opinion on conservation of Ludlow.	0.3										
Output 1.2: Ludlow conservation threats assessed and mapped.											
Action 1.2.1. Carry out grazing mapping and impact assessment.	0.1										
Action 1.2.2 Carry out impact assessment through farming practices	0.1										
Action 1.2.3 carry out impact assessment through resources allocation	0.1										
Objective 2: Conserve Ludlow's habitats based on scientific habitat management.											
Output 2.1. Habitats of Ludlow enhanced through plantation of host and feeding plants											
Action 2.1.1. Upscale the existing Ludlowi host plant nursery to supply enough host and feeding plant seedlings	0.6										
Action 2.1.2: Extend the habitat of Ludlowi through creation/ enrichment plantation of host and feeding plants (5 sites)	1										
Action 2.1.3. Develop Mud Flakes for butterflies in the habitat (5 sites)	0.15										
Output 2.2. Ludlow habitats protected											

Action 2.2.1: Protect prime <i>Bhutanitis ludlowi</i> habitat through natural fencing at some critical habitat with lesser disturbances to other wildlife species	0.9										
Action 2.2.2. Ensure proper management of Waste in the habitat and along the trails											
Output 2.3. Reduce threats through periodic SMART patrol											
Action 2.3.1 Conduct SMART patrol on regular basis	1.0										
Objective 3: Advocate local communities on Ludlow conservation and involve in participatory action planning, implementation and monitoring processes.											
Output 3.1. Awareness on Ludlowi Conservation and stakeholder engagement enhanced											
Action 3.1.1: Develop Information, education and communication materials on <i>Bhutanitis ludlowi</i> for BWS information centre.	0.3										
Action 3.1.2. Produce video documentary of Ludlow life cycle	0.5										
Action 3.1.3: Conduct awareness, advocacy and education programs at all levels of stakeholders (monastic institutions, schools, youths and the general public).	0.3										
Action 3.1.4: Capacity development of park staff on butterfly ecology and survey techniques.	0.8										
Action 3.1.5. Organize butterfly spotting programs for schools and communities.	0.3										
Action 3.1.6: Involve stakeholders in planning, review and evaluation of Ludlowi conservation action plan.	0.5										
Objective 4: Promote community engagement in Ludlow's conservation and support nature-based enterprises, eco-tourism, and other livelihood programs.											
Output 4.1. Strengthened community participation and enhance community resilience through diversification of livelihood activities											
Action 4.1.1. Initiate ecotourism focused on Ludlowi to enhance awareness and conservation	3										
Action 4.1.2: Provide trainings to local youths on nature guide	0.3										
Action 4.1.3 Procure equipment for local nature guides	1.5										
Activity 4.1.4. Support alternative fencing materials to minimize felling and collection of host tree for <i>Aristolochia griffithii</i>	5										
Total Budget Estimate (Nu in Million)	18.1										

4.3. Monitoring and evaluation

Based on the annual operation plan developed by the respective field offices, the implementation of the plan will be jointly monitored annually by the Nature Conservation Division (NCD) and funding agencies. A mid-term review of the plan will be carried out towards the end of the fifth year of plan implementation (Table 04). The logical framework and implementation plan will be used for monitoring and evaluation by using indicators provided.

Table 04. Monitoring and Evaluation

Objective/ Output and Action	Objectively verifiable indicators	Mean of verification	Target	Implementing Agencies	
				Lead	Collaborator
Objective 1: Generate baseline data on Ludlow’s abundance, conservation threats and social demographic status within the Ludlow’s potential habitats.					
Output 1.1: Ludlow habitats mapped and assessed.					
Action 1.1.1: Carry out distribution and habitat mapping	All possible habitats mapped	Report	Map and report produced by 2024	BWS	UWIFoRT
Action 1.1.2: Conduct ecological assessment of Ludlowi	Assessment done	Report and recommendation	Assessment report produced	BWS	UWIFoRT
Action 1.1.3. Carry out annual population monitoring of Ludlowi and its habitat in identified permanent transect.	Monitoring report and data	Final report	Annual report produced	BWS	
Action 1.1.4 Map all the communities in the habitat and study their knowledge and opinion on conservation of LudlowI.	Social survey conducted	Report	Report produced	BWS	
Output 1.2.					
Action 1.2: Ludlow conservation threats assessed and mapped.	Assessment done	Report and recommendation	Assessment report produced	BWS	UWIFoRT
Action 1.2.2 Carry out impact assessment through farming practices	Assessment done	Report and recommendation	Assessment report produced	BWS	UWIFoRT
Action 1.2.3 carry out impact assessment through resources allocation	Assessment done	Report and recommendation	Assessment report produced	BWS	UWIFoRT

Objective 2: Conserve Ludlow's habitats based on scientific habitat management.					
Output 2.1. Habitats of Ludlow enhanced through plantation of host and feeding plants					
Action 2.1.1. Upscale the existing Ludlowi host plant nursery to supply enough host and feeding plant seedlings	Upscaled nursery	Structure and seedling production	1 Nursery	BWS	CFMGs
Action 2.1.2: Extend the habitat of Ludlowi through creation/ enrichment plantation of host and feeding plants (5 sites)	Created/ enriched ludlowi habitat	Physical achievement	5 sites	BWS	CFMGs
Action 2.1.3. Develop Mud Flakes for butterflies in the habitat (5 sites)	Mud flakes developed	Physical achievement	5 sites	BWS	
Output 2.2. Ludlow habitats protected					
Action 2.2.1: Protect prime <i>Bhutanitis ludlowi</i> habitat through natural fencing at some critical habitat with lesser disturbances to other wildlife species	Fenced the sites	Physical achievement	3 sites	BWS	CFMGs
Action 2.2.2. Ensure proper management of Waste in the habitat and along the trails	Mass cleaning conducted	Quarterly report	2 trails	BWS	Local community
Output 2.3. Reduce threats through periodic SMART patrol					
Action 2.3.1 Conduct SMART patrol on regular basis		SMART report	4 times	BWS	

Objective 3: Advocate local communities on Ludlow conservation and involve in participatory action planning, implementation and monitoring processes.					
Output 3.1. Awareness on Ludlowi Conservation and stakeholder engagement enhanced					
Action 3.1.1: Develop Information, education and communication materials on <i>Bhutanitis Ludlow</i> for BWS information centre.	Developed information and communication materials	Printed and displayed		BWS	
Action 3.1.2. Produce video documentary of Ludlow life cycle	Short documentary produced covering four seasons	Ludlowi documentary	15 min document video produced	NCD	BWS
Action 3.1.3: Conduct awareness, advocacy and education programs at all levels of stakeholders (monastic institutions, schools, youths and the general public).	create awareness on Ludlowi	Awareness report produced	School-4, local community -100 and other stakeholders - 30	BWS	
Action 3.1.4: Capacity development of park staff on butterfly ecology and survey techniques.	BWS staff knowledge enhanced	Training conducted	29 staffs	BWS	UWIFoRT
Action 3.1.5. Organize butterfly spotting programs for schools and communities.	Organized 4 times butterfly watching	Report produced	4 schools	BWS	
Action 3.1.6: Involve stakeholders in planning, review and evaluation of Ludlowi conservation action plan.	Involved stakeholders for 3 times	Review report produced	3 times	BWS	UWIFoRT, NCD

Objective 4: Promote community engagement in Ludlow's conservation and support nature-based enterprises, eco-tourism, and other livelihood programs.					
Output 4.1. Strengthened community participation in Ludlow conservation					
Action 4.1.1. Initiate ecotourism focused on Ludlowi to enhance awareness and conservation	Improved trails, amenities, resting points,	eco-trails and amenities developed	2 sites	BWS	
Action 4.1.2: Provide trainings to local youths on nature guide	Local Youths trained for 2 times	Report produced	trained 20 youths	BWS	UWIFoRT
Action 4.1.3 Procure equipment for local nature guides	Guiding equipment procured	Lists of equipment	By 2nd year	BWS	
Activity 4.1.4. Support alternative fencing materials to minimize felling and collection of host tree for <i>Aristolochia griffithii</i>	Effective fencing material supported	physical achievement	By 9th year	BWS	LG/DAO

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